



SpaceX all-civilian crew finalised

A four-person SpaceX civilian crew, sponsored by billionaire Jared Isaacman, has been finalised. <http://digit.in/apr21-47>



Synthetic cells

Scientists have developed a synthetic cell that grows and divides uniformly; can be used to detect disease. <http://digit.in/apr21-48>

Quantum Materials

A new class of materials with strange and useful properties are being investigated for future applications

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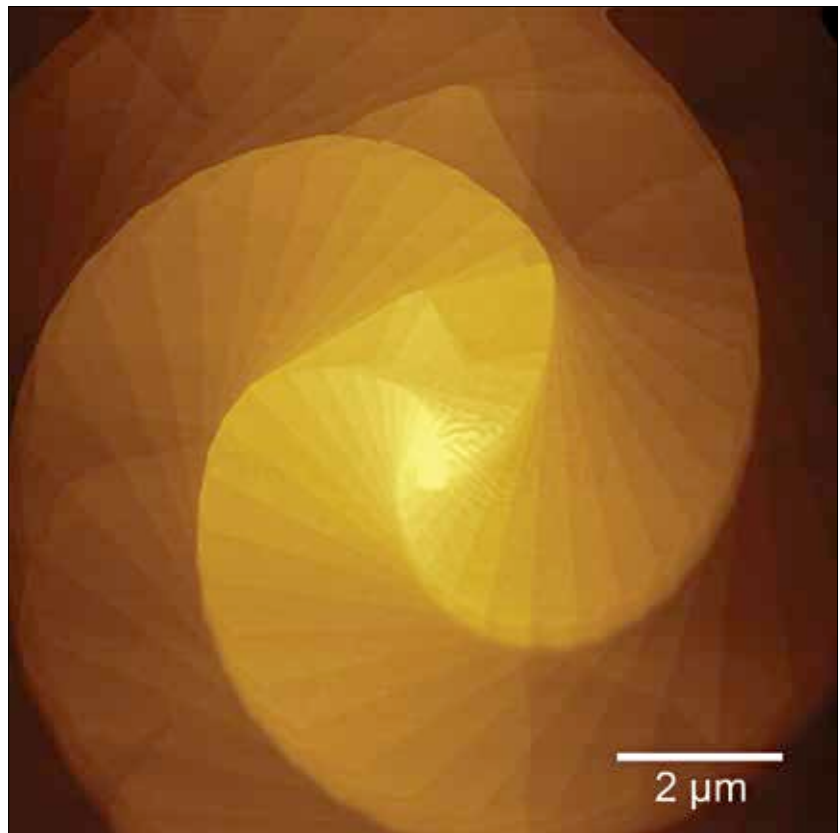
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uantum materials are a diverse variety of materials whose properties cannot be completely explained

by classical physics or even low level quantum mechanics. These materials have bizarre properties and can include topological crystalline insulators, exotic superconductors and strange metals. However, in order to better study quantum materials, researchers need to use quantum materials. Research into quantum materials may allow humans to transport energy over vast distances with minimal or no loss, and reduce the energy consumed in transport.

MOS2

The miniaturisation of electronics has packed in an exponential number of components since the silicon revolution in the 60s (read all about this in the dmystify), but Moore's Law seems to be reaching its limits. One of the active fields of research are memristors, electrical components that can modify the resistance between its two terminals. These components have been implemented for some time now, but researchers do not yet fully understand how they work, and there



A twisting spiral with tunable superconductive properties grown by depositing sheets of material a single atom in thickness over a substrate

Credit: Song Jin Lab

is doubt that a theoretically perfect meristor can even exist in reality. Researchers from the University of Texas had developed a memory storage device which was a type of memristor, called an atomrister, which used a thin film of molybdenum disulfide (MoS₂), a single atom in thickness. MOS₂ belongs to a group of materials known as transition metal chalcogenides, which have interesting electrical properties and optoelectronic behavior.

The same team in November last year reduced the area of the memory unit as well, bringing down the

memory function to a single atom.

The device uses defects or holes in the nanomaterial, that is a single atom in thickness, which is occupied by a single metal atom, which transfers some of its conducting properties to the nanomaterial, the change allowing for the memory storage. Memory devices using the new technology can potentially store 25 terabits per square centimeter, a memory density that is about a hundred times higher than the Flash memory currently available in the market. Pani Varanasi, program manager for the US Army Research